

AP Stats Lesson 2.1 Homework

Answer Key

1.
 - a. 400 freshmen selected
 - b. 3,000 incoming freshmen
2. Conducting a census of the U.S. population requires a lot of planning and staff members and is therefore very expensive (roughly \$13.7 billion). It is also challenging to contact every member of the population, and therefore takes a lot of time.
3.
 - a. Voluntary response. Random chance was not used to select individuals as part of the sample. Instead, there was an open invitation and participants self-selected.
 - b. Those people who feel strongly about protecting the environment would be more likely to volunteer to participate in the survey than those who don't feel strongly about protecting the environment.
 - c. Higher
4.
 - a. Convenience sample. The entrepreneurs did not use random chance to select individuals. Instead, they simply contacted people that were easy to reach.
 - b. Coffee shops provide space and good internet access, so people at coffee shops are more likely to be remote workers than people who are not at coffee shops.
 - c. Higher
5.
 - a. Label each customer with a number from 1 to 91. Use a random number generator to generate 10 numbers between 1 and 91 with no repeats. Contact each customer that corresponds to the numbers that are generated.
 - b. Write each customer's full name on equal sized pieces of paper. Put the papers in a hat and mix them well. Randomly select 10 pieces of paper from the hat (without replacement) and contact each customer that is listed on the pieces of paper selected.
6. D
7. B